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```
clc;  
clear;  
close all;
```

EV BATTERY SAFETY MONITORING

```
% Example safety limits  
max_voltage = 54;           % V  
min_voltage = 39;           % V  
max_current = 50;           % A  
max_temperature = 60;       % deg C  
max_SOC = 100;              % %  
min_SOC = 0;                % %
```

Test Scenarios

[Voltage, Current, Temperature, SOC]

```
scenarios = [  
    52, 40, 35, 85;           % Normal  
    56, 40, 35, 85;           % Over-voltage  
    35, 40, 35, 50;           % Under-voltage  
    52, 65, 35, 85;           % Over-current  
    52, 40, 65, 85;           % Over-temperature  
    54, 40, 35, 105           % Overcharge  
];  
  
scenario_names = {  
    'Normal'  
    'Over-Voltage'  
    'Under-Voltage'  
    'Over-Current'  
    'Over-Temperature'
```

```
        'Overcharge'  
};
```

Result Arrays

```
safety_status = zeros(1, size(scenarios, 1));  
  
over_voltage = zeros(1, size(scenarios, 1));  
under_voltage = zeros(1, size(scenarios, 1));  
over_current = zeros(1, size(scenarios, 1));  
over_temperature = zeros(1, size(scenarios, 1));  
overcharge = zeros(1, size(scenarios, 1));
```

Safety Checking

```
fprintf('=====\n');  
fprintf('          EV BATTERY SAFETY TEST\n');  
fprintf('=====\n\n');
```

```
for i = 1:size(scenarios, 1)  
    voltage = scenarios(i, 1);  
    current = scenarios(i, 2);  
    temperature = scenarios(i, 3);  
    SOC = scenarios(i, 4);  
  
    safe = true;  
  
    fprintf('Scenario %d: %s\n', i, scenario_names{i});
```

Scenario 1: Normal

Scenario 2: Over-Voltage

Scenario 3: Under-Voltage

Scenario 4: Over-Current

Scenario 5: Over-Temperature

Scenario 6: Overcharge

Voltage Check

```
if voltage > max_voltage  
    fprintf(' WARNING: Over-voltage detected\n');  
    over_voltage(i) = 1;  
    safe = false;  
  
elseif voltage < min_voltage  
    fprintf(' WARNING: Under-voltage detected\n');  
    under_voltage(i) = 1;  
    safe = false;
```

```
else
    fprintf(' Voltage: SAFE\n');
end
```

Voltage: SAFE

WARNING: Over-voltage detected

WARNING: Under-voltage detected

Voltage: SAFE

Voltage: SAFE

Voltage: SAFE

Current Check

```
if abs(current) > max_current
    fprintf(' WARNING: Over-current detected\n');
    over_current(i) = 1;
    safe = false;

else
    fprintf(' Current: SAFE\n');
end
```

Current: SAFE

Current: SAFE

Current: SAFE

WARNING: Over-current detected

Current: SAFE

Current: SAFE

Temperature Check

```
if temperature > max_temperature
    fprintf(' WARNING: Over-temperature detected\n');
    over_temperature(i) = 1;
    safe = false;

else
    fprintf(' Temperature: SAFE\n');
end
```

Temperature: SAFE

Temperature: SAFE

Temperature: SAFE

Temperature: SAFE

WARNING: Over-temperature detected

Temperature: SAFE

SOC Check

```
if SOC > max_SOC
    fprintf(' WARNING: Overcharge detected\n');
    overcharge(i) = 1;
    safe = false;

elseif SOC < min_SOC
    fprintf(' WARNING: Invalid SOC detected\n');
    safe = false;

else
    fprintf(' SOC: SAFE\n');
end
```

SOC: SAFE

SOC: SAFE

SOC: SAFE

SOC: SAFE

SOC: SAFE

WARNING: Overcharge detected

Overall Status

```
if safe
    fprintf(' >>> Overall Status: SAFE\n\n');
    safety_status(i) = 1;
else
    fprintf(' >>> Overall Status: UNSAFE\n\n');
    safety_status(i) = 0;
end
```

>>> Overall Status: SAFE

>>> Overall Status: UNSAFE

>>> Overall Status: UNSAFE

```

>>> Overall Status: UNSAFE

>>> Overall Status: UNSAFE

>>> Overall Status: UNSAFE

end

fprintf('=====\n');
fprintf('          TEST COMPLETED\n');
fprintf('=====\n');

=====
          EV BATTERY SAFETY TEST
=====

=====
          TEST COMPLETED
=====

```

GRAPH 1: Overall Safety Status

```

figure;

bar(safety_status);

grid on;

xticks(1:length(scenario_names));
xticklabels(scenario_names);

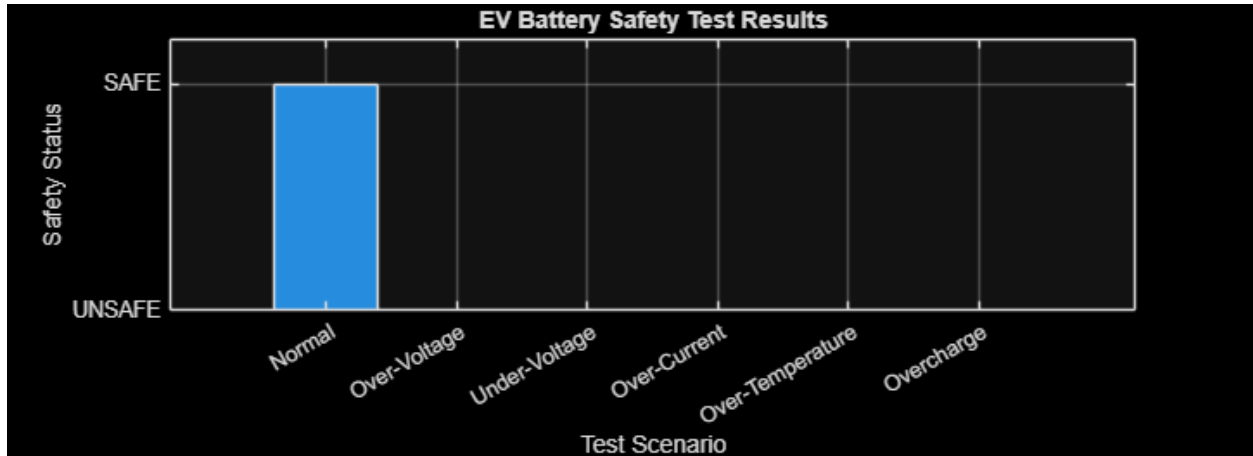
xlabel('Test Scenario');
ylabel('Safety Status');

title('EV Battery Safety Test Results');

ylim([0 1.2]);

yticks([0 1]);
yticklabels({'UNSAFE', 'SAFE'});

```



GRAPH 2: Fault Identification

```
figure;

fault_matrix = [
    over_voltage;
    under_voltage;
    over_current;
    over_temperature;
    overcharge
];

bar(fault_matrix');

grid on;

xticks(1:length(scenario_names));
xticklabels(scenario_names);

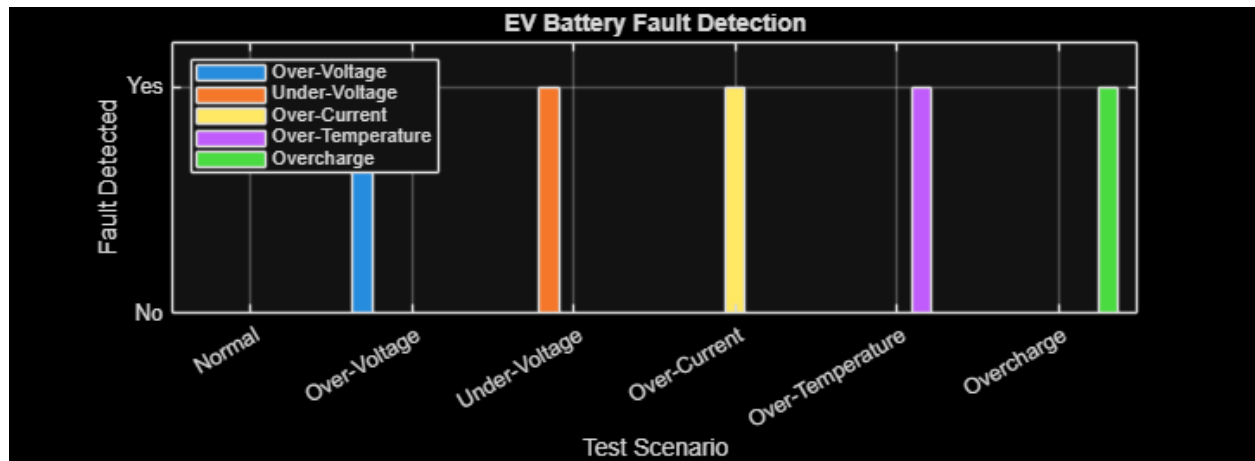
xlabel('Test Scenario');
ylabel('Fault Detected');

title('EV Battery Fault Detection');

legend('Over-Voltage', ...
       'Under-Voltage', ...
       'Over-Current', ...
       'Over-Temperature', ...
       'Overcharge', ...
       'Location', 'northwest');

ylim([0 1.2]);

yticks([0 1]);
yticklabels({'No', 'Yes'});
```



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